



AI Literacy Training

Chapter 1: Architecture

What a Large Language Model (LLM) is and how we can understand them better.

AI as a Family of (Non-Analytical) Methods

Artificial Intelligence (AI)

(e.g., a Roomba, a spam filter, Pokemon battle AI)

Computational architectures that perform **goal-directed inference** via heuristic search or statistical learning.

Machine Learning (ML)

(e.g., Linear Regression, Weather Forecasting)

Statistical algorithms that perform **function approximation** through empirical data.

Neural Network (NN)

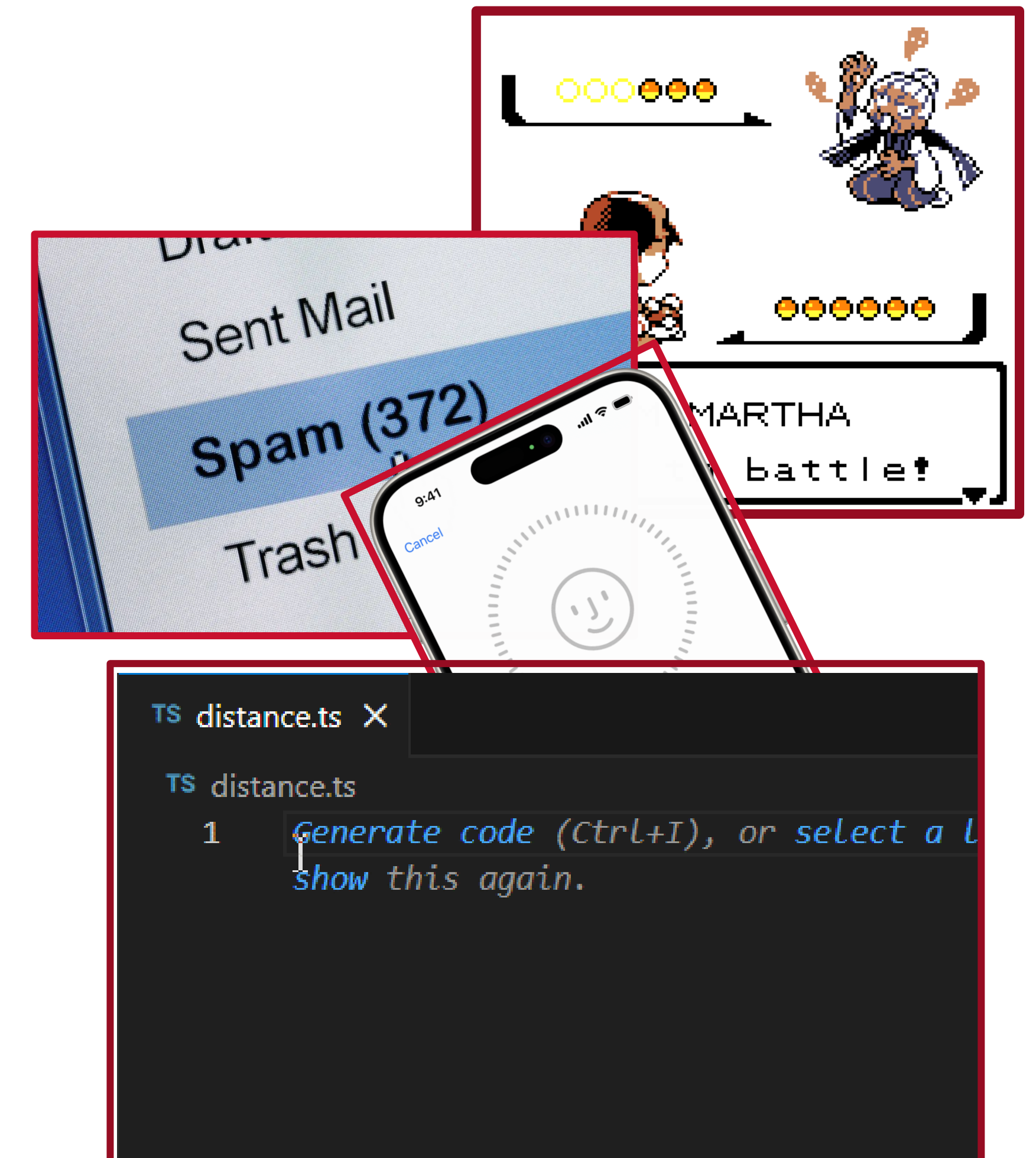
(e.g., Google Translate, Facial Recognition)

Non-linear ML models built from **layered weighted inputs** and perform function fitting through **non-linear activation**.

Large Language Model (LLM)

(e.g., Chatbots, Copilots)

Multi-layered neural network trained via self-supervision and human alignment, to predict and generate **contextual sequences of text**.



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TOKENS & CONTEXT



LIVE DEMONSTRATION



THE MODEL IN LLM



NETWORK & OUTPUT



SUMMARY & TIPS

Contextual Text Prediction $\neq$ Ground Truth

A solid aluminum cube rests on the flat horizontal floor of a tank filled with water. The interface between the bottom face of the cube and the tank floor is a perfect, watertight seal such that no liquid water or moisture can seep or exist underneath the bottom face of the cube. Does the net hydrostatic vertical force exerted by the water on the aluminum cube act upward, downward, or is it zero? Explain the exact fluid pressure integration. Be very concise. Answer within 1-2 sentences.

“A solid aluminum cube rests on the flat horizontal floor of a tank filled with water. The interface between the bottom face of the cube and the tank floor is a perfect, watertight seal such that no liquid water or moisture can seep or exist underneath the bottom face of the cube.”

First Agent's answer: *“Net hydrostatic force is zero.”*

Second Agent's answer: *“Net hydrostatic force is downward.”*

The model stumbled because it relied on memorized text patterns rather than calculating the physical forces from first principles.

❖ **Root Cause:** The model predicted the most common textbook phrasing, instead of calculating the actual physics.

❖ **Inherent LLM Risk:** Probabilistic pattern-matching makes their responses sound entirely confident even when they are **fundamentally wrong**.

Claude is AI and can make mistakes. Please double-check responses.

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LIVE DEMONSTRATION

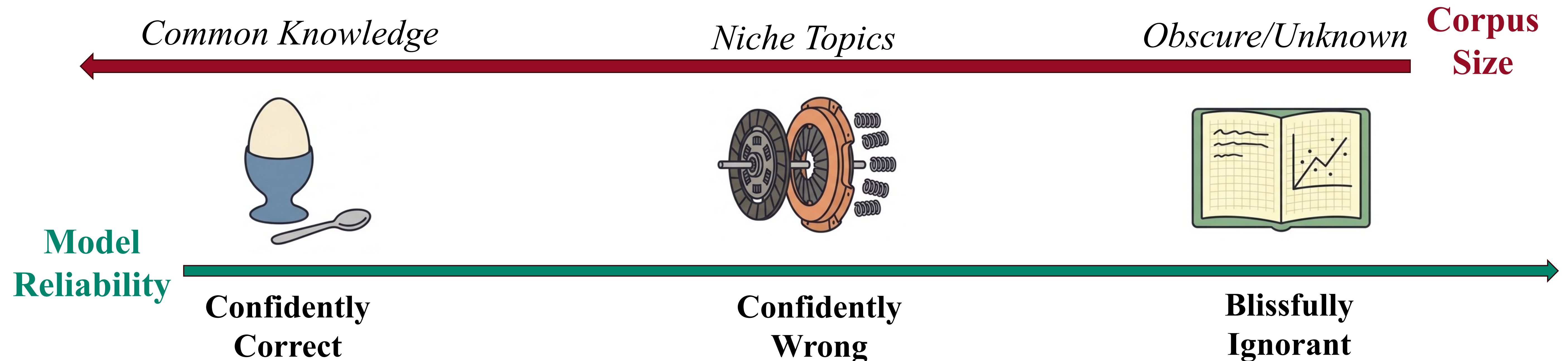
THE MODEL IN LLM

NETWORK & OUTPUT

SUMMARY & TIPS

Confidently Wrong or Blissfully Ignorant?

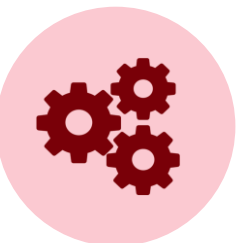
It depends on the topic's availability on the **Corpus** and the model's **Reliability**.



- ❑ **Corpus:** The general and unstructured collection of training material used to develop an LLM's baseline knowledge (i.e., training data).
- ❑ **Reliability:** A dimension of AI trustworthiness representing the property of an AI system to consistently maintain intended behavior and satisfy performance criteria under expected operational distributions.



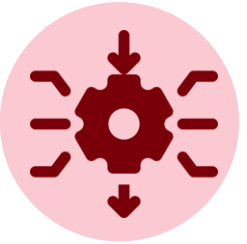
More Terminology

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❑ **Training**^(Empirical Risk Minimization-ERM): The computational phase where parameters are iteratively updated via backpropagation and Stochastic Gradient Descent (SGD) to minimize an empirical loss function. *(Hint: Deep Learning → Gradient Based Optimization).*
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❑ **Knowledge Cutoff** ^(Temporal Distribution Boundary) : The last date of training corpus, defining the temporal limit of the model's parametric knowledge. *(Hint: Ask the model, “What’s your knowledge cutoff date?”).*
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❑ **Prompt**^(Conditioning Sequence): The initial input token sequence, $\mathbf{x}_{\{1:k\}}$, passed to the model to condition its autoregressive predictive distribution, $P(\mathbf{x}_{\{k+1:j\}}|\mathbf{x}_{\{1:k\}})$, determining the probability distribution over all possible subsequent tokens. *(Hint: Session → Conditional Distribution of Current & Previous Prompts).*
- 

❑ **Context Window**^(Sequence Length) : The maximum number of tokens, $\mathbf{x}_{\{1:k_{max}\}}$, the model can process in a single pass. Limited by VRAM in runtime. Compute grows non-linearly $O(k_{max}^2)$. *(Hint: Higher k_{max} → Lower Response Speed).*
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❑ **Inference** ^(Autoregressive Decoding) : The model execution phase where an input token sequence, $\mathbf{x}_{\{1:k\}}$, is evaluated across the static weights (parameters) to evaluate $P(\mathbf{x}_{\{k+1:j\}}|\mathbf{x}_{\{1:k\}})$, generating sequential output tokens. *(Hint: Word by word prediction.)*



Large (Corpus) ✓ Language (Text) ✓ Model ?

Model is the *package* that consists of **the parameters** and a **program** that processes said parameters.

Parameters

(a long array of numbers)

[-0.0412 0.9037 0.1188 -1.2740
0.0006 0.4451 -0.3319 0.7702
....]

Program

(the processing algorithm)

1. Read Parameters into **Memory**
2. **Network** multiplies against token vectors
3. Output: Probability distribution per **Token**

❑ Parameters: Also called weights. Represents model size (e.g., Qwen 32B). **Set in training.**

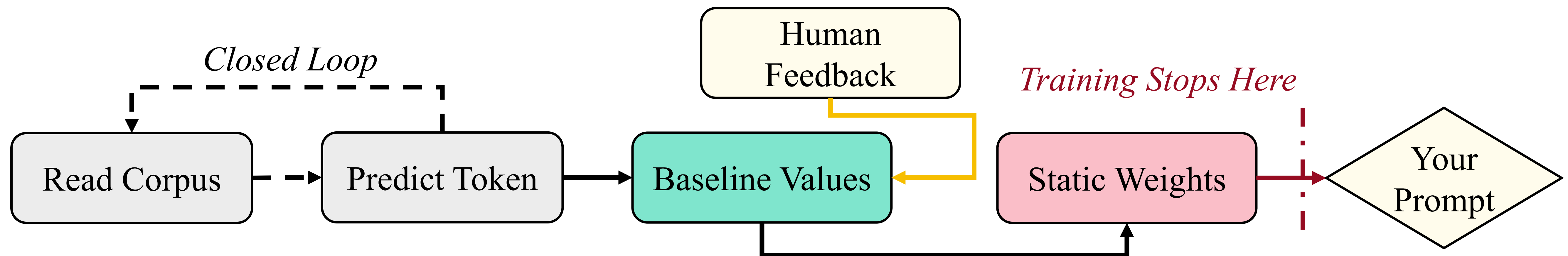
❑ **Token: The fundamental unit of text (word or piece of word) processed by the model.**

❑ Network: The **layered architecture** performing weighted sums and non-linear transformations.



Setting the Parameters (Static Weights)

The list of parameters is **calculated and set** during model training, *in two stages*:



1. **Unsupervised Training:** The model processes the massive corpus (trillions of unlabeled tokens) with the *objective of correctly predicting the next word* (token) in a sentence. This step sets the **baseline values** of all the parameters.
2. **Human Alignment:** Human experts rank and rate model responses against the same prompt. *The model adjusts the weights to **increase the probability of high-quality responses***. This is done in a fraction of the training data (<1%).

**Based on OpenAI's tokenizer (GPT-4o-o200k_base). Different models → Different tokenizer.*

Input & Output: What is a Token? (1/2)

“The fundamental unit of text (word or piece of word) processed by the model.”

What your prompt is: |

What the model sees:*

stirrups → **stir** + **r** + **ups**?

- ☐ Occurrence frequency in the corpus is **too low**.
- ☐ ‘stir’ appears **frequently**. (e.g., *stirrer*, *stir-fry*)
- ☐ ‘rups’ isn’t a **valid** word, but ‘ups’ is.
- ☐ ‘r’ isn’t a word, but it is a **raw byte** (i.e., 81).

Then, do typos matter?

- ☐ Yes... *but not that much*.
- ☐ People make them all the time → **Corpus Vocabulary**.
- ☐ ‘Reinforcement’ → `_reinforcement` → 117657
- ☐ ‘Reinforcement’ → `_rein | for | c | ment` → 17855, 1938, 66, 508



**Based on OpenAI's tokenizer (GPT-4o-o200k_base). Different models → Different tokenizer.*

Input & Output: What is a Token? (2/2)

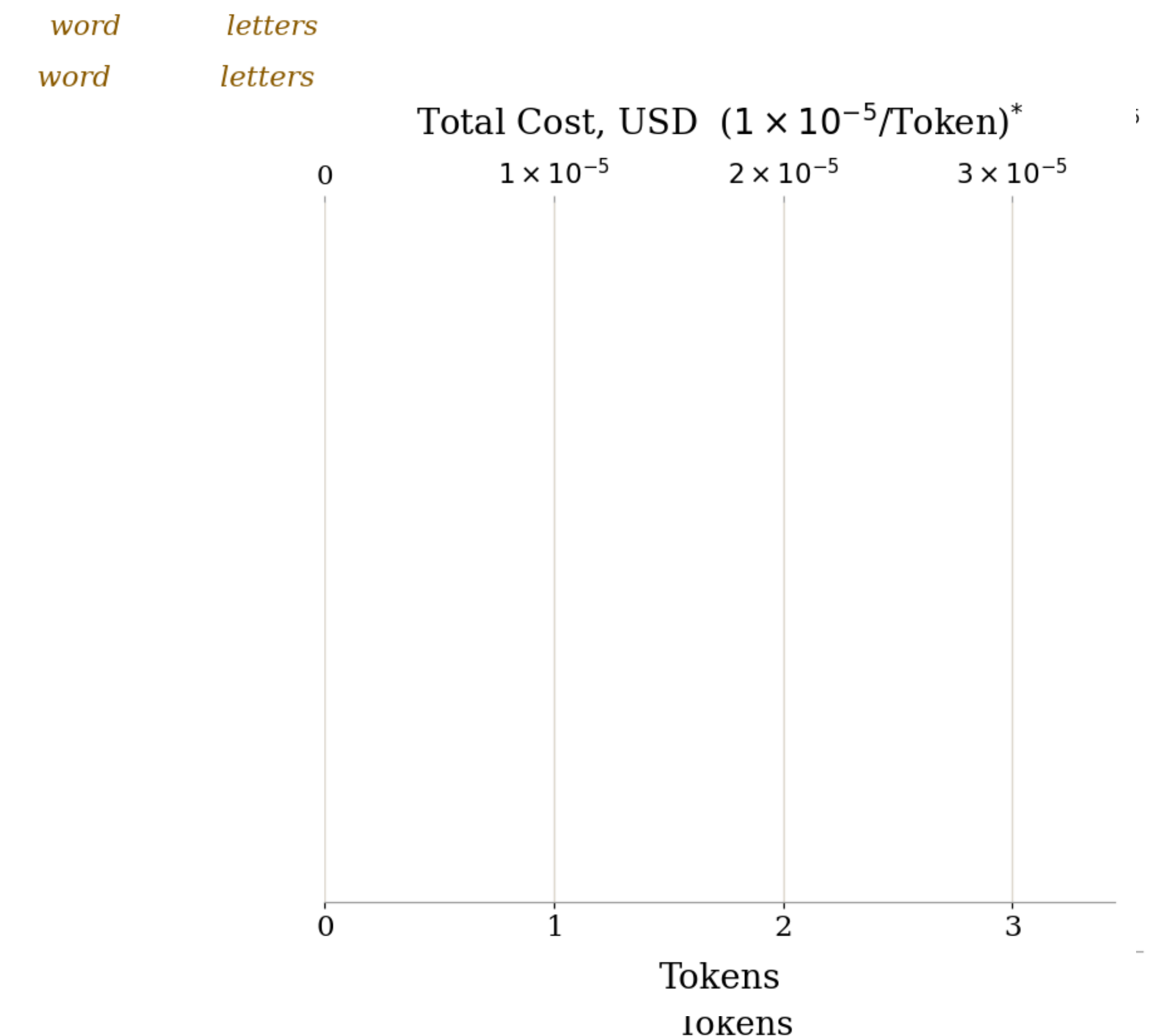
Token ID ≠ Token Count.

Which one do I care about?

- ❑ Token ID → Internal Bookkeeping.
- ❑ **Token Count → Costs \$ and time.**

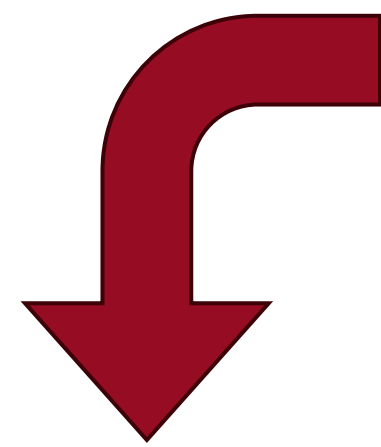
What controls the token count?

- ❑ **Both** ‘**the**’ and ‘**characteristics**’ → 1 token*.
- ❑ **Both** ‘**permeability**’ and ‘**geotechnical**’ → 12 letters.
 - ❑ **But** ‘**permeability**’ → 1 token
 - ‘**geotechnical**’ → 3 tokens
- ❑ **Deciding Factors:** Training, Frequency, and Typos.

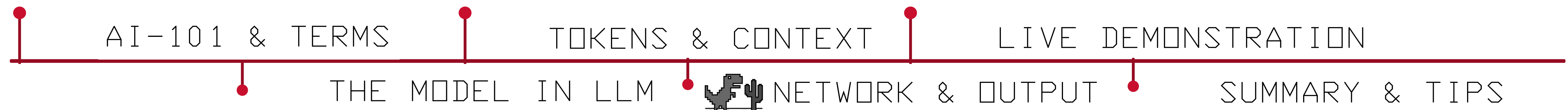


Neural Network: Setting Model's Capabilities

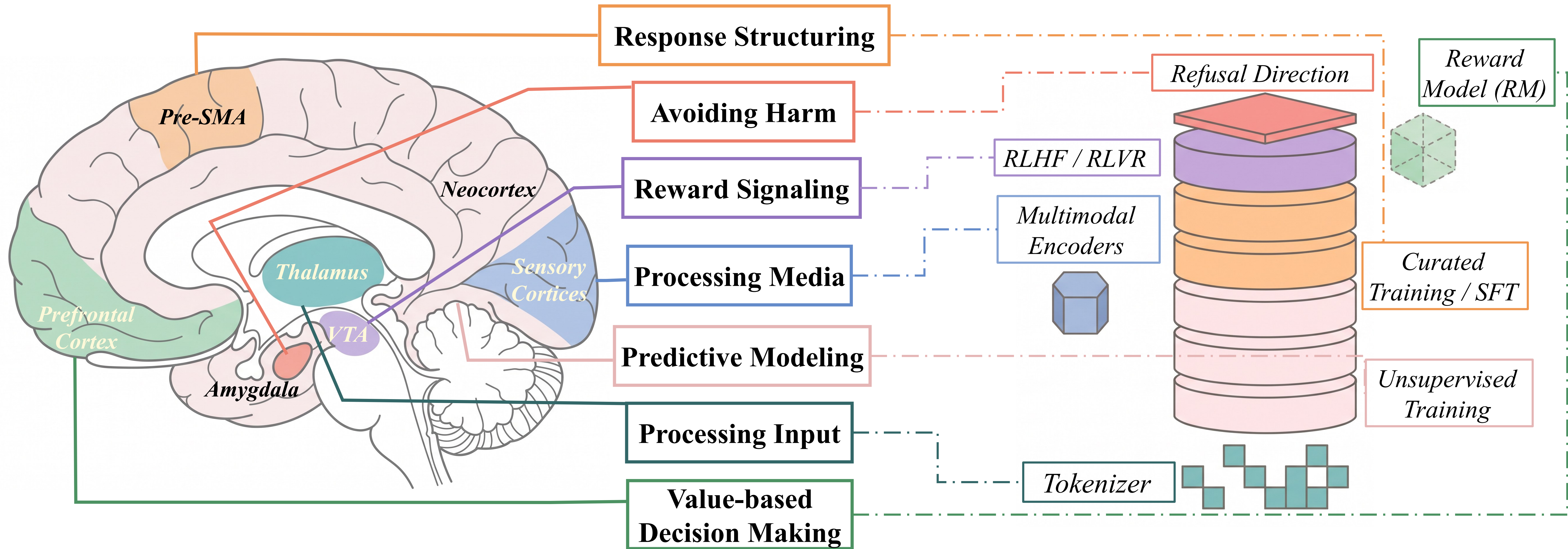
- ❑ Frontier LLMs have many capabilities. How do they acquire them?
- ❑ How are they able to understand text, images, and sound at the same time?
- ❑ What gives a model its 'personality' and 'morality'?
- ❑ How does a model decide how to structure its response?



- ❖ Multi-stage training includes **specialized networks** (e.g., encoders, reward models) with **independent weights**.
 - ❖ The *deployed model* consolidates these learned behaviors into **a unified parameter space**.
 - ❖ A model can enhance its capabilities by **talking to other algorithms** (i.e., MCPs & APIs).
- ❑ **Application Programming Interface (API):** A defined interface through which interactions occur between a software system and other software systems, specifying the operations available, the format of requests and responses, and the semantics of each.
 - ❑ **Model Context Protocol (MCP):** An open protocol standardizing how external tools (e.g., Gmail) and data sources (e.g., GitHub) are made available to an LLM's operating environment.



Neural Network: Layering them in **Training**



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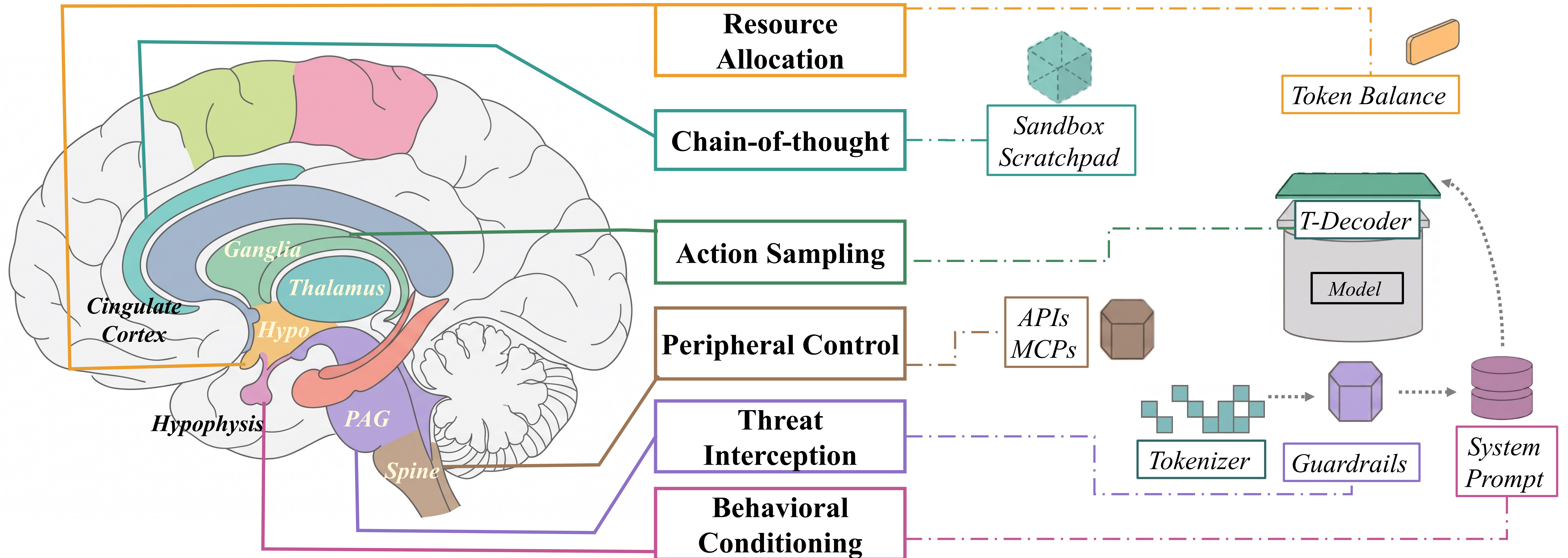
THE MODEL IN LLM



NETWORK & OUTPUT

SUMMARY & TIPS

Neural Network: Starting a Session



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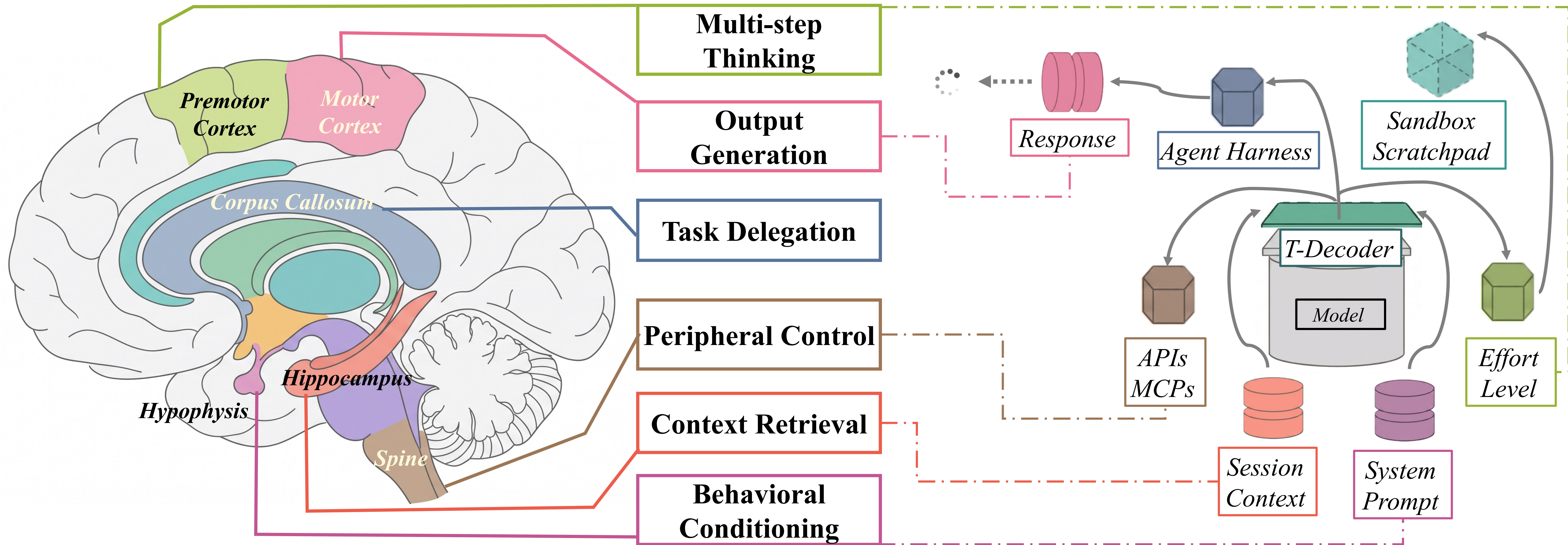
THE MODEL IN LLM



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SUMMARY & TIPS

Neural Network: Processing the Prompt



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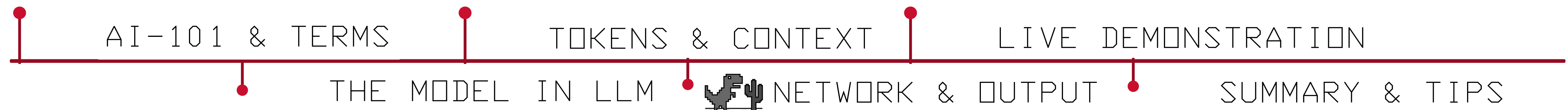
SUMMARY & TIPS

Neural Network: Decision-making Math

Let's define a **program** that calculates “**what would come next**”, using the **model's weights** and the **prompt**.

$$\mathbf{z} = \mathbf{W} \cdot \mathbf{x}_{1:k} + \mathbf{b}$$

The main slides should show focus on showing the equations of how the LLM network is made and how each ‘network’ or thing discussed in the brain slides connects. Mathematically. Lastly, we need to show how everything CLICKs together. What a logit is, randomness, seeds, temperature



Neural Network: Introducing Variance

Equations 1-2 of HOW the output is produced (temperature/seeds) slides and then A WORKED example to show how neural network-> deterministic but IT shouldn't be



Case Study 1: Setting up a LLM

You must find the value, y , for any given value of x .

However,

❖ You have *no idea* what the symbolic equation is. ❖ Your *only feedback* comes from a game of **hot** and **cold**.

□ You *assume* a **program** that calculates “**what would come next**”, using the **model’s weights** and the **prompt**.

$$\mathbf{z} = \mathbf{W} \cdot \mathbf{x}_{1:k} + \mathbf{b}$$

$$P(\mathbf{x}_{k+1} = \mathbf{v} \mid \mathbf{x}_{1:k}; \theta) = \frac{e^{z_v}}{\sum_{j \in \mathcal{V}} e^{z_j}}$$

The *weights and bias* are random numbers before training.

□ You *define* a **loss function** (residual calculation).

□ You *evaluate* $\mathcal{L}(\theta)$ over the corpus, $\mathcal{D}$, ($i = 1 \dots n$).

$$\mathcal{L}(\theta) = -\frac{1}{n} \sum_{i=1}^n \log P(\mathbf{x}_{k+1}^{(i)} \mid \mathbf{x}_{1:k}^{(i)}; \theta)$$

$$\mathcal{D} = \left\{ \left(\mathbf{x}_{1:k}^{(i)}, \mathbf{x}_{k+1}^{(i)} \right) \right\}_{i=1}^n$$

Example:

$$\mathcal{D} = \{ ("1", "3"), ("2", "5"), ("London", "2"), \dots \}$$



Conclusion

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THE MODEL IN LLM

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SUMMARY & TIPS